

LoRaWAN Will Help Us Stay Informed About Our Infrastructure In Real Time

➤ Since its inception from TATA Steel, the Jamshedpur Utilities and Services Company (JUSCO) carries out all functions of urban development and infrastructure maintenance in the Steel City of India. In conversation with Paromik Chakraborty of Electronics For You, Ashish Mathur, MD, JUSCO, explains how the latest technology enables them to shape a smarter, greener and sustainable Jamshedpur

Q. What smart technologies does JUSCO use to provide an efficient service to the citizens?

A. We generate a single consolidated bill for all utilities like electricity, water, municipal contributions and property taxes for the convenience of citizens.

The whole city is geographic information system (GIS) mapped, which gives us real-time information on any part of Jamshedpur at any time. Power distribution is completely managed through a supervisory control and data acquisition (SCADA) system. As a result, compared to the average 22 per cent power distribution loss throughout



Ashish Mathur, MD, JUSCO

the country, the total power loss in Jamshedpur stands at just 4 per cent.

The JUSCO Sahyog Kendra gives us access to a rich repository of data, which we use to gain insight into higher complaint areas.

We use connected technology to monitor which trash bin is getting full and should be taken out—saving a lot of manual effort and time in trash disposal and hygiene maintenance. Smart lighting is also a major action area to conserve electricity.

In short, our focus is conservation of resources using technology in the smartest way possible.

Q. How does LoRaWAN help in integrated city management?

A. LoRaWAN will help us stay informed about our infrastructure in real time with the desired efficiency. For example, if we want to test the quality of the city-wide drinking water online, instead of sampling and testing in labs, sensors across the water outlets will measure water quality parameters and send this data in real time to the integrated command centre of our office through the LoRaWAN network. So we will be able to act immediately if required.

This technology will help us in numerous other applications too. For instance, sensors on sewer drain covers will monitor the flow of liquid levels inside. When the liquid level crosses a certain threshold, a signal to our command centre will inform

us about a possible choking of the drain, enabling us to take action before it overflows.

Jamshedpur is already completely under the LoRaWAN network. The challenge is to develop products and sensors that can talk to this network and relay data to our integrated command centre at the other end of the network. A lot of research is under progress, and it may take 6-8 months for mass deployment of these solutions.

Q. What are its advantages over other networks?

A. A major parameter for choosing the LoRaWAN was its impact on battery life. Sensors and devices used across the city for major utilities will provide real-time online measurement, where functional loss due to drained battery is not feasible. Additionally, these devices will be used in areas where charging points will be obscure—for example, within pipelines, where devices may need to run on battery power for up to a year. So we had to look for a technology that uses little battery power, and LoRaWAN suited our requirement.

In addition, we wanted to limit our capex and cost of usage. Here again, LoRaWAN was a good fit. Ease of use was another deciding factor.

Q. How can India achieve the vision of sustainable cities?

A. A big transformation may start from the thought process of bringing all the urban utilities like electricity, water distribution and public health under one umbrella. This will help to speed up services, put standardisations and accountabilities in place, and achieve an overall efficiency.

We need to solve local issues at local level itself. The different stakeholders—municipal corporations, citizens, communities and service providers alike—need to come together and work as a team to make the whole process cost- and time-effective. **EFY**

If you want to share such case studies with EFY readers, please contact us at smartworld@efy.in